

Linear algebra – 6

Linear algebra in combinatorics

Homework

12 сентября 2026 года

Theoretical minimum. The dimension method; characteristic vectors over $\mathbb{R}$ and over $\mathbb{F}_2$; the rank bound $\text{rank}(A + B) \leq \text{rank } A + \text{rank } B$; Oddtown, Eventown and Fisher's inequality; the polynomial method and the Combinatorial Nullstellensatz; the Frankl–Wilson theorem on families with restricted intersections modulo a prime.

1. Prove Fisher's inequality, and deduce the Frankl–Wilson bound: if p is a prime, $|A_i| \equiv \lambda \pmod{p}$ for all i and $|A_i \cap A_j| \not\equiv \lambda \pmod{p}$ for $i \neq j$, then $m \leq \binom{n}{p-1} + \binom{n}{p-2} + \cdots + \binom{n}{0}$.
2. (Graham–Pollak 1971) Prove that if the edges of K_n are partitioned into k complete bipartite graphs, then $k \geq n - 1$. (*Hint: compute the signature of a suitable quadratic form.*)
3. (IMC) Let n be a positive integer and let V be a $(2n - 1)$ -dimensional vector space over the two-element field. Prove that for arbitrary vectors $v_1, \dots, v_{4n-1} \in V$ there are indices $1 \leq i_1 < \cdots < i_{2n} \leq 4n - 1$ with $v_{i_1} + \cdots + v_{i_{2n}} = 0$.
4. (IMC) Let k be a positive integer. Find the smallest n for which there exist nonzero vectors $v_1, \dots, v_k \in \mathbb{R}^n$ such that $v_i \perp v_j$ for every pair of indices with $|i - j| > 1$.
5. (IMC) Consider the $n \times n$ matrix with entries $1, 2, \dots, n^2$ written in order along the rows. We choose n entries, exactly one in each row and each column. What are the possible values of their sum?
6. (Mexmat, 2009) Let $m(k)$ be the maximum number of pairwise non-orthogonal vectors in $\{-1, 0, 1\}^{2k}$ each having exactly k zero coordinates.

- a) Prove that $m(k) \geq 2^{2k-1}$ for odd k .
- b) Prove that $80 \leq m(4) \leq 140$.
7. (VJIMC 2005, cat. II) Let $(a_{ij})_{i,j=1}^n$ be a real matrix with $a_{ii} = 0$ for all i . Prove that there is a set of indices $J \subseteq \{1, \dots, n\}$ such that

$$\sum_{i \in J, j \notin J} a_{ij} + \sum_{i \notin J, j \in J} a_{ij} \geq \frac{1}{2} \sum_{i,j=1}^n a_{ij}.$$

8. (IMC) A sequence $(A_1, \dots, A_k)$ of real $n \times n$ matrices is called *preferred* if $A_i^2 \neq O$ for all i while $A_i A_j = O$ for all $i \neq j$. Show that $k \leq n$ for every preferred sequence, and give an example with $k = n$ for each n .
9. (IMC 2019) Let $x_1, \dots, x_n$ be real numbers and for $I \subseteq \{1, 2, \dots, n\}$ put $s(I) = \sum_{i \in I} x_i$. Assume that the function $I \mapsto s(I)$ takes at least 1.8^n distinct values as I runs over all 2^n subsets of $\{1, 2, \dots, n\}$. Prove that the number of sets $I \subseteq \{1, 2, \dots, n\}$ with $s(I) = 2019$ does not exceed 1.7^n .
10. (Cauchy–Davenport) Prove that $|A + B| \geq \min\{p, |A| + |B| - 1\}$ for non-empty $A, B \subseteq \mathbb{Z}/p\mathbb{Z}$ with p prime, using the Combinatorial Nullstellensatz, and compare the proof with the case $n = 1$ of the Brunn–Minkowski inequality.